

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 10
 - Compactness
 - Metric Spaces

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Warm-up Challenge

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- Lindelöf Covering Theorem

Bolzano-Weierstrass Theorem

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about real numbers.

Theorem (Bolzano-Weierstrass Theorem)

A bounded set $A \subseteq \mathbb{R}^n$ with infinitely many points must have an accumulation point.

Cantor Intersection Theorem

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Suppose that $C_1, C_2, C_3, \dots$ are non-empty closed, bounded sets with $C_{i+1} \subseteq C_i$ for all $i \in I = \mathbb{Z}_+$. Then $\bigcap_{i \in I} C_i$ is also non-empty.

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Therefore the Bolzano-Weierstrass Theorem tells us it has an accumulation point $\vec{x}$.



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Since C_m is closed, it follows $\vec{x} \in C_m$, and thus $\vec{x} \in \bigcap_{i \in I} C_i$. In particular the intersection is non-empty! □

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Since $\vec{x} \in A \subseteq \bigcup_{i \in I} U_i$, we have that $\vec{x} \in U_k$ for some $k \in I$, meaning $\vec{x} \notin C_k$.

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This is a contradiction.



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The value $d(x, y)$ is called a **metric** and describes the “distance” between x and y .

Open balls

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Definition

Let (M, d) be a metric space. The open ball of radius $r > 0$ centered at $x \in M$ is

$$B_M(x; r) = \{y \in M : d(x, y) < r\}.$$

Examples of metrics on $\mathbb{R}^2$

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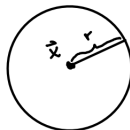
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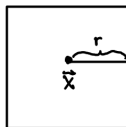
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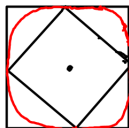
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Challenge!

Let M be a nonempty set and define $d : M \times M \rightarrow \mathbb{R}$ by

$$d_{\text{disc}}(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

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This is called the **discrete metric**.

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Problem

What do the open balls with the discrete metric look like?